An example of an innovative panel survey approach for eliciting guideline panelists' views on the minimally important difference for patients

We describe here the experience of a guideline panel making recommendations for patients with a high risk of myocardial infarction regarding a new treatment to reduce that risk.[1] To interpret whether a certain effect of treatment on myocardial infarction is important or not, the panel required information about the smallest reduction in myocardial infarction that patients would perceive as important (the MID threshold).

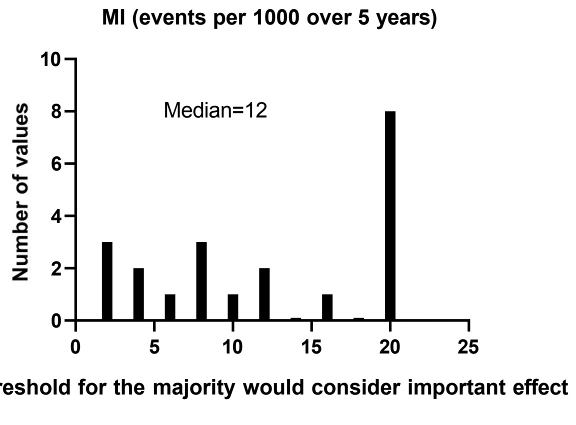
Based on the perceived need for clarification regarding the MID, a team of methodologists designed a survey to elicit guideline panelists' views on the smallest reduction in myocardial infarctions that patients would experience as important.[2]

The survey presented a series of scenarios in which the magnitude of effect on reducing myocardial infarctions varied. The first scenario was "in adults considering the possibility of using the new treatment to reduce the risk of myocardial infarction, the treatment lowers their risk by 1 in 1000 over a period of 5 years". In the following scenarios, the reduction in myocardial infarctions changed to 20, 3, 15, 5, 10, 8 and 12 in 1000 (a ping-pong approach going from one extreme to another, gradually narrowing the differences). Under each scenario, the survey asked panelists to make inferences regarding the proportion of patients who would consider the particular magnitude of effect on myocardial infarction as either important or trivial.

The options include:
- All or almost all would consider this an important effect
- Most would consider this an important effect
- A majority would consider this an important effect
- A majority would consider this a trivial effect
- Most would consider this a trivial effect
- All or almost all would consider this a trivial effect

The guideline panelists reflected on the questions based on the experience of clinician panel members who had engaged in shared decision making with their patients, as well as conversations addressing health care decisions with friends and family.

When a panelist switched the response from "a majority would consider this an important effect" to "a majority would consider this a trivial effect" (or vice versa), the panelist identified a narrow range within which the MID lies. The panel took the average mean of the MID thresholds identified by individual guideline panelists as the MID the panel considered to inform the interpretation of the magnitude of effect. The distribution of results was as follows:



Based on these results the panel adopted a minimal important difference of a reduction in myocardial infarction of 12 in 1,000, the value at which approximately half the panel thought that the majority of patients would consider the effect as important and half would consider the reduction in myocardial infarction unimportant and half unimportant.

The full panel survey is as follows.

Introduction

Based on the last panel survey, we identified the reduction in myocardial infarction as a key desirable outcome. Now we need your help in considering what impact on the outcome that patients would consider important. This will help us in rating precision when deciding on certainty of evidence and will ultimately help in deciding on recommendations for or against the drugs under consideration, and the strength of those recommendations.

We need you to answer a series of questions regarding what patients would consider a trivial or important effect. At this point, the question is abstract in that it isn't tied to the benefits, harms, or burdens of interventions. These judgments are challenging. If you are a clinician, please reflect the question based on your experience in shared decision-making with your patients. If you are attending the panel in the role of patient, please answer the question on the basis of conversations with friends, family, and acquaintances around making health care decisions.

In the following questions, when we say "all or almost all", we mean 90% or more; when we say most, we mean 75% to 90%; and when we say the majority, we mean 50% to 74% (also applicable to other examples in Appendix 3).

In each case, the proportion who think an effect is important will be 100% minus the proportion who think an effect is trivial. For instance, if you choose the option "the majority would consider this a trivial effect" it means that you think that 50% to 74% would think the effect trivial and 26% to 49% would think it is important.

1. In adults considering the possibility of using medication to reduce their risk of myocardial infarction, an intervention lowers their risk by **1 in 1000** (i.e. a decrease in myocardial infarction of 1 in 1,000 patients) over a period of 5 years. Please choose an option that will reflect the proportion of patients who would think this reduction in risk is an (a) important/trivial effect?
   - All or almost all would consider this an important effect
   - Most would consider this an important effect
   - A majority would consider this an important effect
   - A majority would consider this a trivial effect
   - Most would consider this a trivial effect
   - All or almost all would consider this a trivial effect

2. In adults considering the possibility of using medication to reduce their risk of myocardial infarction, an intervention lowers their risk by **20 in 1000** (i.e. a decrease in myocardial infarction of 15 in 1,000 patients) over a period of 5 years. Please choose an option that will reflect the proportion of patients who would think this reduction in risk is an (a) important/trivial effect?
   - All or almost all would consider this an important effect
   - Most would consider this an important effect
   - A majority would consider this an important effect
   - A majority would consider this a trivial effect
   - Most would consider this a trivial effect
   - All or almost all would consider this a trivial effect

3. In adults considering the possibility of using medication to reduce their risk of myocardial infarction, an intervention lowers their risk by **3 in 1000** (i.e. a decrease in myocardial infarction of 15 in 1,000 patients) over a period of 5 years. Please choose an option that will reflect the proportion of patients who would think this reduction in risk is an (a) important/trivial effect?
   - All or almost all would consider this an important effect
   - Most would consider this an important effect
   - A majority would consider this an important effect
   - A majority would consider this a trivial effect
   - Most would consider this a trivial effect
   - All or almost all would consider this a trivial effect

4. In adults considering the possibility of using medication to reduce their risk of myocardial infarction, an intervention lowers their risk by **15 in 1000** (i.e. a decrease in myocardial infarction of 15 in 1,000 patients) over a

period of 5 years. Please choose an option that will reflect the proportion of patients who would think this reduction in risk is an (a) important/trivial effect?

- All or almost all would consider this an important effect
- Most would consider this an important effect
- A majority would consider this an important effect
- A majority would consider this a trivial effect
- Most would consider this a trivial effect
- All or almost all would consider this a trivial effect

5. In adults considering the possibility of using medication to reduce their risk of myocardial infarction, an intervention lowers their risk by **5 in 1000** (i.e. a decrease in myocardial infarction of 15 in 1,000 patients) over a period of 5 years. Please choose an option that will reflect the proportion of patients who would think this reduction in risk is an (a) important/trivial effect?
   - All or almost all would consider this an important effect
   - Most would consider this an important effect
   - A majority would consider this an important effect
   - A majority would consider this a trivial effect
   - Most would consider this a trivial effect
   - All or almost all would consider this a trivial effect

6. In adults considering the possibility of using medication to reduce their risk of myocardial infarction, an intervention lowers their risk by **10 in 1000** (i.e. a decrease in myocardial infarction of 15 in 1,000 patients) over a period of 5 years. Please choose an option that will reflect the proportion of patients who would think this reduction in risk is an (a) important/trivial effect?
   - All or almost all would consider this an important effect
   - Most would consider this an important effect
   - A majority would consider this an important effect
   - A majority would consider this a trivial effect
   - Most would consider this a trivial effect
   - All or almost all would consider this a trivial effect

7. In adults considering the possibility of using medication to reduce their risk of myocardial infarction, an intervention lowers their risk by **8 in 1000** (i.e. a decrease in myocardial infarction of 15 in 1,000 patients) over a period of 5 years. Please choose an option that will reflect the proportion of patients who would think this reduction in risk is an (a) important/trivial effect?
   - All or almost all would consider this an important effect
   - Most would consider this an important effect
   - A majority would consider this an important effect
   - A majority would consider this a trivial effect
   - Most would consider this a trivial effect
   - All or almost all would consider this a trivial effect

8. In adults considering the possibility of using medication to reduce their risk of myocardial infarction, an intervention lowers their risk by **12 in 1000** (i.e. a decrease in myocardial infarction of 15 in 1,000 patients) over a period of 5 years. Please choose an option that will reflect the proportion of patients who would think this reduction in risk is an (a) important/trivial effect?
   - All or almost all would consider this an important effect
   - Most would consider this an important effect
   - A majority would consider this an important effect
   - A majority would consider this a trivial effect
   - Most would consider this a trivial effect
   - All or almost all would consider this a trivial effect

References

1. Hao Q, Aertgeerts B, Guyatt G, et al. PCSK9 inhibitors and ezetimibe for the reduction of cardiovascular events: a clinical practice guideline with risk-stratified recommendations. *Bmj* 2022;377:e069066. doi: 10.1136/bmj-2021-069066 [published Online First: 20220504]
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